PRODUCT USER MANUAL

For Global Ocean – High Resolution SAR Sea Ice Drift

SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_006,
SEAICE_GLO_PHY_L4_MY_011_020

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PUM for SI TAC global ice drift product SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_006 SEAICE_GLO_PHY_L4_MY_011_020	Ref: CMEMS-SI-PUM-011-006-020 Date : November 30th 2023 Issue: 2.14

CHANGE RECORD

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2.7	2016-09-30	I.2	Sentinel-1B added	Roberto Saldo	B. Hackett
2.8	2018-01-25		Rebranded from OSI to SI TAC	C. Wettre	
2.9	2019-04-09	I.1, III.3, IV.2	MOSAIC product contents explained	Roberto Saldo	
2.10	2019-09-18	III.3	MOSAIC product contents explained	Roberto Saldo	C. Derval

2.11	2019-12-06	III.3	MOSAIC product contents explained	Roberto Saldo	
2.12	2020-03-27		MOSAIC product added deformation variables	Roberto Saldo	
2.13	2022-11-30	I.1	Sentinel-1B loss repercussions in summary	Roberto Saldo	
2.14	2023-05-30		New multiyear reprocessed composite dataset: SEAICE_GLO_PHY_L4_MY_011_020	Roberto Saldo	C. Wettre
2.14	30.11.2023	IV	Updated description tables 1 and 2	Roberto Saldo	C. Wettre

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GLOSSARY AND ABBREVIATIONS

TAC	Thematic Assembly Centre
SI	Sea Ice
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
RMS	Root mean square
SDN	SeaDataNet (climatology)
PC	Production Center
PU	Production Unit
DU	Dissemination Unit
ftp	Protocol to download files
OpenDAP	Open-Source Project for a Network Data Access Protocol. Protocol to download subset of data from a n-dimensional gridded dataset (ie: 4 dimensions: lon-lat,depth,time)
Subsetter	CMEMS service tool to download a NetCDF file of a selected geographical box using values of longitude an latitude, and time range

I INTRODUCTION

I.1 Summary

This guide describes the Global Sea Ice Drift data product files from DTU Space, what data services are available to access them, and how to use the files and services.

DTU-GLO-SEAICE_DRIFT-NORTH-L4-NRT-OBS or DTU-GLO-SEAICE_DRIFT-SOUTH-L4-NRT-OBS is a set of gridded ice displacement fields obtained by near real time MCC processing of SAR data.

DTU-GLO-SEAICE_DRIFT-NORTH-L4-MOSAIC-OBS or DTU-GLO-SEAICE_DRIFT-SOUTH-L4-MOSAIC-OBS is a 24hr mean composite of the gridded ice displacement fields.

The multiyear product SEAICE_GLO_PHY_L4_MY_011_020 contains the datasets *cmems_obs-si_glo_phy-drift-north_my_l4_P1D-m* and *cmems_obs-si_glo_phy-drift-south_my_l4_P1D-m*. This is the reprocessed multiyear version of the 24hr composite products.

Note: Data from before August 23, 2017 are given as ice-drift per day (dx and dy) divided by the grid spacing (10 km), so in order to get ice-drift in km/day you should multiply the numbers by the grid spacing. After August 23, 2017, the dx and dy values are given as km/day.

Originally the SAR data used was from Envisat ASAR WSM swath data resampled to 300m resolution. Since the loss of the Envisat satellite in 2012, replacement wide-swath scenes from Radarsat2 have been used. The nominal temporal span between processed swaths is 24 hours, the nominal product grid resolution is 10km.

As of December 15th 2014 the product is processed from Sentinel 1 SAR data resampled to 300m.

As of June 2020 three drift derived deformation variables are included.

The data are disseminated by CMEMS DU via the MET Norway PC, and are freely available under the CMEMS data license.

On the 23rd of December 2021 an anomaly rendered the Sentinel-1B satellite inoperable. Unfortunately the Sentinel-1B was responsible for acquiring SAR data over the Central Arctic and the result is almost no coverage above 80N thus a huge pole-hole in the mosaic product as most of the Sea Ice is located there.

I.2 History of changes

Date	Impact
Since October 2014	Sentinel-1 A data has gradually been incorporated into this product
Since June 2016	Sentinel-1 B data has been incorporated into this product.
Since December 2016	Introduction of the mosaic product
Since July 2020	Deformation variables available in mosaic product.
December 2023	Introduction of the reprocessed multiyear mosaic product

II DOWNLOAD A PRODUCT

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

III FILES NOMENCLATURE

III.1 Nomenclature of files when downloaded through the Copernicus Marine FTP

The nomenclature of the downloaded files differs on the basis of the chosen download mechanism **Copernicus Marine FTP** service.

The file naming convention used for the SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_006 product is:

ice_drift_polstereo_sarbased_POLE_DATE-thedaybefore-DATE.nc

POLE: north | south

DATE:YYYYMMDDhhmmss

Example: ice_drift_polstereo_sarbased_north_20140101062246-20140102151120.nc

The file naming convention is relevant when downloading by **FTP**.

III.2 Nomenclature of files when downloaded through the Copernicus Marine Web Portal Subsetter Service

where:

.datasetname is a character string within one of the following:

- ice_drift_north
- ice_drift_south

. nnnnnn: 6 digit transaction identifier.

.zip: files within the requested time range are packed by zip.

Example: **ice_drift_south_151494.nc**

IV DESCRIPTION OF THE PRODUCT SPECIFICATION

IV.1 General Information

Two different near-real time datasets are available for either poles, the near-real time single scene swath matches the high detail drift set and the 24hr mean composite mosaic.

The composite dataset also exists as a reprocessed version to cover further back in time, since the launch of Sentinel 1A in 2014.

IV.2 High resolution near real time single swath data

Upstream SAR data is re-sampled post-process to 300m resolution before sea ice drift is calculated by using the maximum cross correlation method on single geographically overlapping scenes from consecutive dates

IV.3 High resolution 24 hour mean value composite/multiyear reprocessed

The composite is produced by filtering and calculating a grid of the mean values of the individual single swath data sets normalised to 24 hours.

Data includes time of first and last contributing values used, sample counts, mean and standard deviation values for every grid point.

Note: Data from before August 23, 2017 are given as ice-drift per day (dx and dy) divided by the grid spacing (10 km), so in order to get ice-drift in km/day you should multiply the numbers by the grid spacing. After August 23, 2017, the dx and dy values are given as km/day.

Note: due to a coordinate system confusion with the composite products up until july 2020, line,sample(1,1) > Xc,Yc(0,0) the following needs to be done when reading the data fields. These issues may be corrected in future updates so be aware of the version numbering!

1) data fields are offset by one line and one sample (1,1) that is ((lon[i+1,j+1], lat[i+1,j+1]) corresponds to (xc[i], yc[j]). Data from files marked :product_version = "1.1" and software_version = "v20170823" and possibly older versions, the data fields need to be shifted before use. In python that can be done with a numpy roll command: lon_fix= np.roll(lon, -901) which fixes the (1,1) offset.

2) dX_mean and dY_mean are swapped, that is dX_mean contains the dY_mean values and dY_mean the dX_mean ones. The same for dX_std and dY_std, they need swapping too.

3) dY_mean needs to be inverted

Summary (pseudo python):

lon_corrected= np.roll(lon, -901),

lat_corrected= np.roll(lat, -901)

dX_mean_corrected= np.roll(dY_mean, -901)

*dY_mean_corrected= -1*np.roll(dX_mean, -901)*

dX_std_corrected= np.roll(dY_std, -901)

dY_std_corrected= np.roll(dX_std, -901)

Please do not hesitate to contact the help desk if you have any questions.

Note: As of composite product version 2.0 from July 2020 the above-mentioned coordinate system is correct and no correction necessary.

V DESCRIPTION OF THE PRODUCT SPECIFICATION

Product Lines	SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_006
Geographical coverage	Northern and southern hemispheres
Variables	Displacement in X and Y
Product Type	Near-real Time
Available time series	**From 2010-03-09 ongoing for Northern hemisphere** **From 2010-03-09 ongoing for Southern hemisphere**
Temporal resolution	Depending on SAR data
Target delivery time	Near-real time
Delivery mechanism	FTP
Horizontal resolution	10 km
Number of vertical levels	1
Format	Single swath Netcdf CF1.4, Composite CF 1.6

Table 1: details about SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_006

V.1 Details of datasets

SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_006		
DATASETS	VARIABLES AND UNIT or GEOGRAPHICAL COVERAGE	NAME and **STANDARD_NAME** OF VARIABLES IN THE NETCDF FILE or DOMAIN AREA
DTU-GLO-SEAICE_DRIFT-NORTH-L4-NRT-OBS	Sea ice drift [km]	dX "**sea_ice_x_displacement**" dY "**sea_ice_y_displacement**"
DTU-GLO-SEAICE_DRIFT-SOUTH-L4-NRT-OBS	Sea ice drift [km]	dX "**sea_ice_x_displacement**" dY "**sea_ice_y_displacement**"
DTU-GLO-SEAICE_DRIFT-NORTH-L4-MOSAIC-OBS	Mean sea ice drift [km]**	dX_mean "**sea_ice_x_displacement**" dY_mean "**sea_ice_y_displacement**"
DTU-GLO-SEAICE_DRIFT-SOUTH-L4-MOSAIC-OBS	Mean sea ice drift [km]**	dX_mean "**sea_ice_x_displacement**" dY_mean "**sea_ice_y_displacement**"
DTU-GLO-SEAICE_DRIFT-NORTH-L4-MOSAIC-OBS	Vector deformation, [1/s]	Divergence "flow divergence" Vorticity" flow vorticity" Shear" flow shear"

DTU-GLO-SEAICE_DRIFT-SOUTH-L4-MOSAIC-OBS	Vector deformation, [1/s]	Divergence "flow divergence" Vorticity" flow vorticity" Shear" flow shear"
SEAICE_GLO_PHY_L4_MY_011_020		
cmems_obs-si_glo_phy-drift-north_my_l4_P1D-m	Vector deformation, [1/s]	Divergence "flow divergence" Vorticity" flow vorticity" Shear" flow shear"
cmems_obs-si_glo_phy-drift-south_my_l4_P1D-m	Vector deformation, [1/s]	Divergence "flow divergence" Vorticity" flow vorticity" Shear" flow shear"

Table 2: List of the datasets (column 1), of the variable for each dataset (column 2) and their names in the NetCDF files (column 3) for the SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_006 and SEAICE_GLO_PHY_L4_MY_011_020

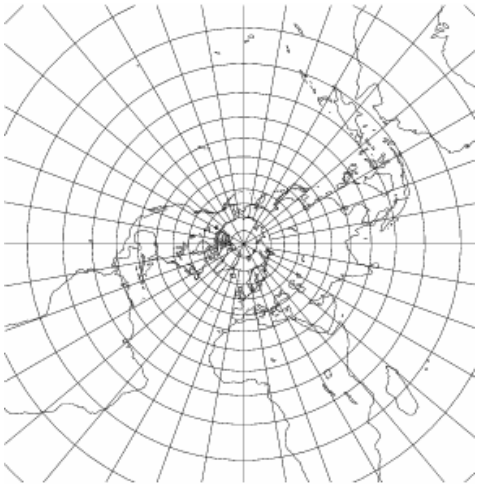
***) Until August 23, 2017, ice drift displacement dx and dy in the MOSAIC products are given as grid cells per day (24 hours). In order to convert to km/day(24h) you should multiply the numbers by the grid spacing (10 km).*

V.2 Grid and Geographical Projection Grid

The product is global, that is both northern and southern hemispheres are covered.

Each dataset has an individual grid depending on the source SAR data and the scarcity of the data.

The grid is set in a Polar Stereographic projection north (proj=stere +R=6370997 +lat_0=90 +lat_ts=70 +lon_0=0) or south (proj=stere +R=6370997 +lat_0=-90 +lat_ts=-70 +lon_0=45) with a spacing of 10.182338 km.

Stereographic projection: The best representation of for the polar regions, north or southern hemisphere.	

V.3 Update Time

Near real time, with every new SAR scene at most 24 hours after SAR data has been acquired.

The composite is updated every 12 hours covering 24 hours with a 12h latency.

V.4 Temporal extent of analysis and forecast stored on delivery mechanism

The product is stored since **2010-03-09.**

VI FILE FORMAT

VI.1 Format and reading software

The products are stored using the NetCDF format. Please follow this link:

[What is the format of Copernicus Marine products ? NetCDF]

VI.2 Structure and semantic of NetCDF NRT files

```
netcdf ice_drift_polstereo_sarbased_north_20131207070428-20131208081639 {
dimensions:
        time = 1 ;
        status = 1 ;
        qos = 1 ;
        nv = 2 ;
        xc = 4 ;
        yc = 4 ;
variables:
        int Polar_Stereographic_Grid ;
                Polar_Stereographic_Grid:grid_mapping_name = "polar_stereographic" ;
                Polar_Stereographic_Grid:straight_vertical_longitude_from_pole = 0.f ;
                Polar_Stereographic_Grid:latitude_of_projection_origin = 90.f ;
                Polar_Stereographic_Grid:standard_parallel = 70.f ;
                Polar_Stereographic_Grid:false_easting = 0.f ;
                Polar_Stereographic_Grid:false_northing = 0.f ;
                Polar_Stereographic_Grid:earth_shape = "spherical" ;
                Polar_Stereographic_Grid:proj4_string = "+proj=stere +R=6370997 +lat_0=90 +lat_ts=70 +lon_0=0" ;
        double time(time) ;
                time:long_name = "reference time of product" ;
                time:standard_name = "time" ;
                time:units = "seconds since 2000-01-01 00:00:00 UTC" ;
                time:calendar = "standard" ;
                time:axis = "T" ;
                time:bounds = "time_bnds" ;
        double time_bnds(time, nv) ;
                time_bnds:units = "seconds since 2000-01-01 00:00:00 UTC" ;
```

```
short status_flag(status) ;
        status_flag:long_name = "quality control indicator" ;
        status_flag:flag_values = 0s, 1s, 2s, 3s ;
        status_flag:flag_meanings = "nominal_quality low_quality rejected_by_filter no_input_data" ;
double qos_bnds(time, nv) ;
        qos_bnds:long_name = "validation period" ;
        qos_bnds:units = "seconds since 2000-01-01 00:00:00 UTC" ;
short qos_sample_count(qos) ;
        qos_sample_count:long_name = "validation statistics sample count" ;
        qos_sample_count:units = "count" ;
double qos_mean_dX(qos) ;
        qos_mean_dX:long_name = "mean of difference against GPS bouys X axis" ;
        qos_mean_dX:units = "meter" ;
double qos_mean_dY(qos) ;
        qos_mean_dY:long_name = "mean of difference against GPS bouys Y axis" ;
        qos_mean_dY:units = "meter" ;
double qos_stddev_dX(qos) ;
        qos_stddev_dX:long_name = "standard deviation of difference against GPS bouys X axis" ;
        qos_stddev_dX:units = "meter" ;
double qos_stddev_dY(qos) ;
        qos_stddev_dY:long_name = "standard deviation of difference against GPS bouys Y axis" ;
        qos_stddev_dY:units = "meter" ;
double xc(xc) ;
        xc:axis = "X" ;
        xc:long_name = "x coordinate of projection (eastings)" ;
        xc:units = "km" ;
        xc:standard_name = "projection_x_coordinate" ;
        xc:grid_spacing = "10.182338 km" ;
double yc(yc) ;
        yc:axis = "Y" ;
        yc:long_name = "y coordinate of projection (northings)" ;
        yc:units = "km" ;
        yc:standard_name = "projection_y_coordinate" ;
        yc:grid_spacing = "10.182338 km" ;
float lat(yc, xc) ;
        lat:long_name = "latitude coordinate" ;
        lat:standard_name = "latitude" ;
        lat:units = "degrees_north" ;
float lon(yc, xc) ;
```

```
                lon:long_name = "longitude coordinate" ;
                lon:standard_name = "longitude" ;
                lon:units = "degrees_east" ;
        float dX(time, yc, xc) ;
                dX:long_name = "component of the displacement along the x axis of the grid" ;
                dX:standard_name = "sea_ice_x_displacement" ;
                dX:units = "km" ;
                dX:grid_mapping = "Polar_Stereographic_Grid" ;
                dX:coordinates = "lat lon" ;
        float dY(time, yc, xc) ;
                dY:long_name = "component of the displacement along the y axis of the grid" ;
                dY:standard_name = "sea_ice_y_displacement" ;
                dY:units = "km" ;
                dY:grid_mapping = "Polar_Stereographic_Grid" ;
                dY:coordinates = "lat lon" ;
// global attributes:
                :Conventions = "CF-1.4" ;
                :title = "Arctic Sea Ice Drift, L4 SAR, 10km daily" ;
                :product_name = "SAR highres ice drift" ;
                :product_status = "preoperational" ;
                :abstract = "Gridded ice displacement fields obtained by near real time MCC processing of Synthetic Aperture Radar scenes resampled to 300m resolution.The nominal temporal span between processed swaths is 24hours,the nominal product grid resolution is a 10km." ;
                :topiccategory = "Oceans ClimatologyMeteorologyAtmosphere" ;
                :keywords = "Sea Ice Motion,Sea Ice,Oceanography,Meteorology,Climate,Remote Sensing" ;
                :activity_type = "space borne instrument" ;
                :area = "northern hemisphere" ;
                :start_date = "2013-12-07 07:04:28 UTC" ;
                :stop_date = "2013-12-08 08:16:39 UTC" ;
                :PI_name = "Roberto Saldo" ;
                :distribution_statement = "free" ;
                :references = "www.seaice.dk" ;
                :history = "2013-12-10 10:46:08 UTC created" ;
                :netcdf_version_id = "3.6.2" ;
                :product_version = "1.0" ;
                :software_version = "1.4" ;
                :institution = "DTU Space - Technical University of Denmark" ;
                :contact = "rs@space.dtu.dk" ;
data:
 Polar_Stereographic_Grid = _ ;
```

```
time = 439805799 ;
time_bnds =
 439715068, 439805799 ;
status_flag = 0 ;
qos_bnds =
 436579200, 439171199 ;
qos_sample_count = 1664 ;
qos_mean_dX = -26.09 ;
qos_mean_dY = -40.71 ;
qos_stddev_dX = 353.35 ;
qos_stddev_dY = 349.36 ;
xc = -143.449, -133.26666, -123.08432, -112.90198 ;
yc = -1205.76368, -1215.94602, -1226.12836, -1236.3107 ;
lat =
 78.77634, 78.787, _, _,
 78.68348, 78.69405, 78.70385, 78.71288,
 78.59062, 78.60111, 78.61083, 78.61978,
 _, _, _, 78.52669 ;
lon =
 -6.78456, -6.307, _, _,
 -6.72827, -6.25461, -5.78009, -5.30477,
 -6.6729, -6.20307, -5.73241, -5.26096,
 _, _, _, -5.21787 ;
dX =
 -2.8718, -2.86125, _, _,
 -3.08324, -3.081, -3.27404, -3.06802,
 -3.70931, -3.49615, -3.28157, -3.07512,
 _, _, _, -3.49635 ;
dY =
 -11.28148, -11.29544, _, _,
 -11.04023, -11.49359, -11.59835, -11.85409,
 -10.75006, -11.45132, -11.60128, -11.8569,
 _, _, _, -11.82116 ;
```

VI.3 Structure and semantic of NetCDF MOSAIC files

```
        netcdf ice_drift_mosaic_polstereo_sarbased_north_20200229000000-20200301000000 {
dimensions:
        time = 1 ;
```

```
        xc = 900 ;
        yc = 900 ;
variables:
        int Polar_Stereographic_Grid ;
                Polar_Stereographic_Grid:grid_mapping_name = "polar_stereographic" ;
                Polar_Stereographic_Grid:straight_vertical_longitude_from_pole = 0.f ;
                Polar_Stereographic_Grid:latitude_of_projection_origin = 90.f ;
                Polar_Stereographic_Grid:standard_parallel = 70.f ;
                Polar_Stereographic_Grid:false_easting = 0.f ;
                Polar_Stereographic_Grid:false_northing = 0.f ;
                Polar_Stereographic_Grid:earth_shape = "spherical" ;
                Polar_Stereographic_Grid:proj4_string = "+proj=stere +R=6370997 +lat_0=90 +lat_ts=70 +lon_0=0" ;
        double time(time) ;
                time:long_name = "reference time of product" ;
                time:standard_name = "time" ;
                time:units = "seconds since 2000-01-01 00:00:00 UTC" ;
                time:calendar = "standard" ;
                time:axis = "T" ;
        double time_period_start(time) ;
                time_period_start:long_name = "sample period start time" ;
                time_period_start:units = "seconds since 2000-01-01 00:00:00 UTC" ;
        double time_period_end(time) ;
                time_period_end:long_name = "sample period end time" ;
                time_period_end:units = "seconds since 2000-01-01 00:00:00 UTC" ;
        short status_flag(time) ;
                status_flag:long_name = "quality control indicator" ;
                status_flag:flag_values = 0s, 1s, 2s, 3s ;
                status_flag:flag_meanings = "nominal_quality low_quality rejected_by_filter no_input_data" ;
        short qos_sample_count(time) ;
                qos_sample_count:long_name = "validation statistics sample count" ;
                qos_sample_count:units = "count" ;
        double qos_mean_dX(time) ;
                qos_mean_dX:long_name = "mean of difference against GPS bouys X axis" ;
                qos_mean_dX:units = "meter" ;
        double qos_bnds_from_time(time) ;
                qos_bnds_from_time:long_name = "validation period start time" ;
                qos_bnds_from_time:units = "seconds since 2000-01-01 00:00:00 UTC" ;
        double qos_bnds_to_time(time) ;
                qos_bnds_to_time:long_name = "validation period end time" ;
```

```
        qos_bnds_to_time:units = "seconds since 2000-01-01 00:00:00 UTC" ;
double qos_mean_dY(time) ;
        qos_mean_dY:long_name = "mean of difference against GPS bouys Y axis" ;
        qos_mean_dY:units = "meter" ;
double qos_stddev_dX(time) ;
        qos_stddev_dX:long_name = "standard deviation of difference against GPS bouys X axis" ;
        qos_stddev_dX:units = "meter" ;
double qos_stddev_dY(time) ;
        qos_stddev_dY:long_name = "standard deviation of difference against GPS bouys Y axis" ;
        qos_stddev_dY:units = "meter" ;
double xc(xc) ;
        xc:axis = "X" ;
        xc:long_name = "x coordinate of projection (eastings)" ;
        xc:units = "km" ;
        xc:standard_name = "projection_x_coordinate" ;
double yc(yc) ;
        yc:axis = "Y" ;
        yc:long_name = "y coordinate of projection (northings)" ;
        yc:units = "km" ;
        yc:standard_name = "projection_y_coordinate" ;
float lat(yc, xc) ;
        lat:long_name = "latitude coordinate" ;
        lat:standard_name = "latitude" ;
        lat:units = "degrees_north" ;
        lat:valid_min = 0.f ;
        lat:valid_max = 90.f ;
float lon(yc, xc) ;
        lon:long_name = "longitude coordinate" ;
        lon:standard_name = "longitude" ;
        lon:units = "degrees_east" ;
        lon:valid_min = -180.f ;
        lon:valid_max = 180.f ;
float dX_mean(time, yc, xc) ;
        dX_mean:standard_name = "sea_ice_x_displacement" ;
        dX_mean:long_name = "mean component of the displacement along the x axis of the grid" ;
        dX_mean:units = "km" ;
        dX_mean:_FillValue = -9999.f ;
        dX_mean:grid_mapping = "Polar_Stereographic_Grid" ;
        dX_mean:coordinates = "xc yc" ;
```

```
            dX_mean:cell_methods = "area: mean" ;
            dX_mean:valid_min = -250.f ;
            dX_mean:valid_max = 250.f ;
float dY_mean(time, yc, xc) ;
            dY_mean:standard_name = "sea_ice_y_displacement" ;
            dY_mean:long_name = "mean component of the displacement along the y axis of the grid" ;
            dY_mean:units = "km" ;
            dY_mean:_FillValue = -9999.f ;
            dY_mean:grid_mapping = "Polar_Stereographic_Grid" ;
            dY_mean:coordinates = "xc yc" ;
            dY_mean:cell_methods = "area: mean" ;
            dY_mean:valid_min = -250.f ;
            dY_mean:valid_max = 250.f ;
float dX_std(time, yc, xc) ;
            dX_std:long_name = "sea_ice_x_displacement_standard_deviation" ;
            dX_std:units = "km" ;
            dX_std:_FillValue = -9999.f ;
            dX_std:grid_mapping = "Polar_Stereographic_Grid" ;
            dX_std:coordinates = "xc yc" ;
            dX_std:cell_methods = "area: standard_deviation" ;
float dY_std(time, yc, xc) ;
            dY_std:long_name = "sea_ice_y_displacement_standard_deviation" ;
            dY_std:units = "km" ;
            dY_std:_FillValue = -9999.f ;
            dY_std:grid_mapping = "Polar_Stereographic_Grid" ;
            dY_std:coordinates = "xc yc" ;
            dY_std:cell_methods = "area: standard_deviation" ;
float dXY_count(time, yc, xc) ;
            dXY_count:long_name = "number of samples per value" ;
            dXY_count:_FillValue = -9999.f ;
            dXY_count:grid_mapping = "Polar_Stereographic_Grid" ;
            dXY_count:coordinates = "xc yc" ;
            dXY_count:units = "count" ;
float divergence(time, yc, xc) ;
            divergence:long_name = "flow divergence" ;
            divergence:_FillValue = -9999.f ;
            divergence:grid_mapping = "Polar_Stereographic_Grid" ;
            divergence:coordinates = "xc yc" ;
            divergence:units = "1/s" ;
```

```
                divergence:valid_min = -1.f ;
                divergence:valid_max = 1.f ;
        float vorticity(time, yc, xc) ;
                vorticity:long_name = "flow vorticity" ;
                vorticity:_FillValue = -9999.f ;
                vorticity:grid_mapping = "Polar_Stereographic_Grid" ;
                vorticity:coordinates = "xc yc" ;
                vorticity:units = "1/s" ;
                vorticity:valid_min = -1.f ;
                vorticity:valid_max = 1.f ;
        float shear(time, yc, xc) ;
                shear:long_name = "flow shear" ;
                shear:_FillValue = -9999.f ;
                shear:grid_mapping = "Polar_Stereographic_Grid" ;
                shear:coordinates = "xc yc" ;
                shear:units = "1/s" ;
                shear:valid_min = -1.f ;
                shear:valid_max = 1.f ;
        double time_first(time, yc, xc) ;
                time_first:long_name = "time of earliest observed sample" ;
                time_first:units = "seconds since 2000-01-01 00:00:00 UTC" ;
                time_first:_FillValue = -9999. ;
                time_first:grid_mapping = "Polar_Stereographic_Grid" ;
                time_first:coordinates = "xc yc" ;
        double time_last(time, yc, xc) ;
                time_last:long_name = "time of latest observed sample" ;
                time_last:units = "seconds since 2000-01-01 00:00:00 UTC" ;
                time_last:_FillValue = -9999. ;
                time_last:grid_mapping = "Polar_Stereographic_Grid" ;
                time_last:coordinates = "xc yc" ;

// global attributes:
                :Conventions = "CF-1.6" ;
                :title = "Arctic Sea Ice Drift, L4 SAR, 10km daily" ;
                :product_name = "SAR highres ice drift composite" ;
                :product_status = "preoperational" ;
                :abstract = "Mean/standard deviation and divergence/vorticity/shear of gridded normalised ice displacement fields obtained by near real time MCC processing of Synthetic Aperture Radar scenes resampled to 300m resolution. The nominal temporal span between processed swaths is 24hours the nominal product grid resolution is 10km." ;
                :topiccategory = "Oceans ClimatologyMeteorologyAtmosphere" ;
```

:keywords = "Sea Ice Motion,Sea Ice,Oceanography,Meteorology,Climate,Remote Sensing" ;
:activity_type = "space borne instrument" ;
:area = "northern hemisphere" ;
:start_date = "2020-02-29 00:00:00 UTC" ;
:stop_date = "2020-03-01 00:00:00 UTC" ;
:PI_name = "Roberto Saldo" ;
:distribution_statement = "free" ;
:references = "www.seaice.dk" ;
:history = "2020-03-24 07:59:39 UTC created" ;
:netcdf_version_id = "4.6.0" ;
:product_version = "2.0" ;
:software_version = "v20200323" ;
:institution = "DTU Space - Technical University of Denmark" ;
:contact = "rs@space.dtu.dk" ;
}

VIIVALIDATION AND ASSESSMENT

Validation is carried out routinely every 3 months (quarterly) against high quality GPS positions from Woods Hole ITPs (http://www.whoi.edu/page.do?pid=20781).

Validation reports are available at http://www.seaice.dk/CMEMS/validation

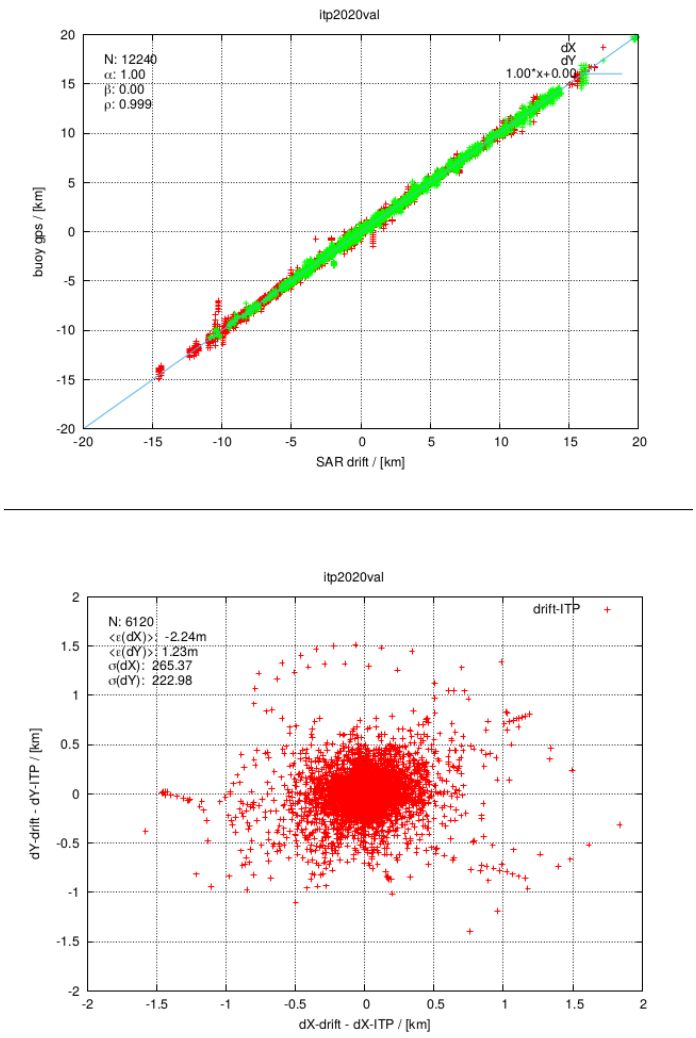


Figure 1 Example of validation scatterplot, comparison of drift against buoy GPS